PARSA NEDAEI

Electrical Engineer
(Telecommunications)

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WHO AM I?

Electrical Engineering graduate with major in ***Telecommunications***, from University of Isfahan and MSc student of Photonics with specialization in ***Opto-electronics*** and, ***Quantum Computing and Information Theory***. Also have practical experience in ***communication systems and quantum information simulations*** and, ***network programming***. My goal is to obtain a research or industrial position that will allow me to utilize my multidisciplinary knowledge in the areas of ***information theory***, ***cryptography***, ***communication networks***, ***quantum communications*** and, ***related topics***. I am eager to contribute to the advancement of modern technologies and knowledge transfer to the next generation in an innovative and academic environment.

- Rust
- C Programming
- Assembly
- Linux
- Bash
- Julia
- Python
- MATLAB

TOOLS STACK

EDUCATION

2024 – Present

Master's Degree - Photonics Engineering — **Shahid Beheshti University**
Supervisor: Dr. M. Ghanaatshoar (*m-ghanaat@sbu.ac.ir*)
Description: Research and learning ***quantum computing***, and ***quantum******classic information theory*** followed by working on applications of semiconductor devices in Quantum communications and computing. Also, designing, simulating, and building Thin-Film transistors with optimizations and developments in device parameters, for ***telecommunication application***.

2020 – 2024

Bachelor's Degree - Electrical Engineering — **University of Isfahan**
Supervisor: Dr. H. Karimi Alavijeh (*hrkarimial@gmail.com*)
Description: Studied *Electrical Engineering* with major in ***Telecommunications***, followed by focus on both analog and digital aspects such as ***Information Theory***, ***RF Circuit Design and Analysis***, ***Optical Communications***, ***Wireless*** and ***Digital Communications***.

EXPERIENCE

Oct 2022 – Jan 2023

Teacher Assistant - **Signals & Systems** — **Faculty of Electrical Engineering, University of Isfahan**
Lecturer: Dr. Negin Sayyaf
Responsibilities: Ensure that students, have at least acceptable understanding of the concepts, teached in lecture. Also designing final project in processing audio signals, and analysis of the effects of filtering, downsampling, and upsampling.
`MATLAB / Mathematics / Python`

Oct 2023 – Jan 2024

Teacher Assistant - **Engineering Electromagnetics** — **Faculty of Electrical Engineering, University of Isfahan**
Lecturer: Dr. Hassan Zamani
Responsibilities: Designing final project, problem sets, and holding troubleshooting sessions.
`COMSOL / MATLAB`

Feb 2024 – May 2024 | *Teacher Assistant* - **Signals & Systems** | **Faculty of Electrical Engineering, University of Isfahan**

Lecturer: Dr. Farzad Parvaresh

Responsibilities: Designing final projects, problem sets, holding troubleshooting sessions, and teaching MATLAB simulations via both online and offline sessions.

`MATLAB / Mathematics / Python`

Feb 2024 – May 2024 | *Laboratory Assistant* - **Laboratory of RF Circuits** | **Faculty of Electrical Engineering, University of Isfahan**

Principal Investigator: Dr. Mohsen Mivehchi

Description: Within this project, we must design a receiver circuit for aviation band signals. This project was meant to be used for new major of avionics, for their laboratory sessions. Within this circuit, several key components such as Low-Noise amplifier, mixer, ceramic filter and ... where used.

Responsibilities: Measurement and quality testing of the designed circuit. Using different instruments of electrical measurements. Also designing PCB and schematic of circuit, with elements in question.

`Altium Desing / RF Electronics / ADS`

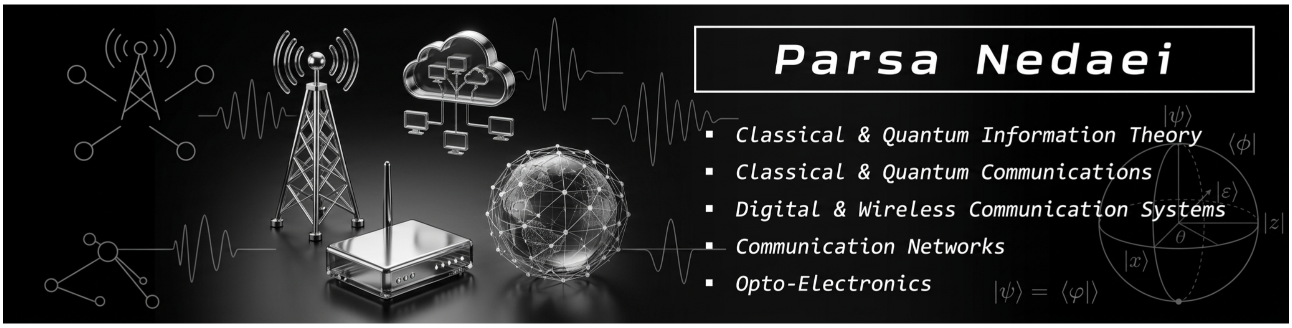
June 2024 – Aug 2024 | *R&D* - **Fiber Laser Tuning Techniques** | **Faculty of Electrical Engineering, University of Isfahan**

Principal Investigator: Dr. Hamidreza Karimi Alavijeh

Description: This project was done by me and colleague, Mrs. A. Mohammad Hoseini, which turned into our final project of bachelor's degree. We used optical circuit elements, and a AZO-DR1 microwire attached to a tapered fiber, to tune the EDF fiber wavelength via UV exposure, for Optical Telecommunication applications. Range of $10\mu m$ of tunibility achieved under linearly polarized near-UV exposure parallel to the position of microwire.

Responsibilities: Assembly, measurement, and quality testing of the optical circuit via different scenarios of UV exposure, in both amplitude and polarization.

`Lumerical Interconnect / COMSOL / Spectrum Analysis / Photonics / Laser Physics`



INTERESTS

- **Classical & Quantum Information Theory**
  Channel Coding (e.g., Polar Codes), Quantum Error Correction Codes (QECC), Quantum Shannon Theory, Rate-Distortion Theory.
- **Communication Networks**
  Next Generation Networks (6G), Network Slicing, Cloud-native Networks, Multi-access Edge Computing (MEC), Resource Allocation Optimization, Software-Defined Networking (SDN).
- **Cryptography & Security**
  Post-Quantum Cryptography (PQC), Physical Layer Security, Quantum Key Distribution (QKD), Zero-Knowledge Proofs.
- **Wireless Communications**
  Massive MIMO Systems, Reconfigurable Intelligent Surfaces (RIS), Millimeter-Wave (mmWave) Communication, Multiple-Access Systems.
- **Quantum Communications**
  Quantum Key Distribution (QKD) Protocols, Entanglement Distribution in Networks, Quantum Repeaters, Quantum Sensing.

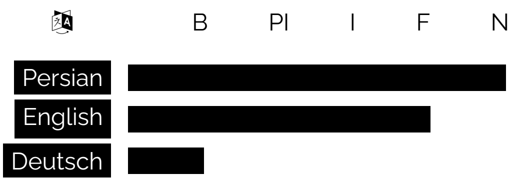
CURRENT RESEARCHS & STUDIES

- Research and learning ***Information Theory***, in both classical and quantum aspects.
- Research and learning ***Digital Communications***; i.e different types of modulations, encoding, and applications.
- Research, coding, and learning ***Communication Networks***; Layers, protocol, and applications.
- Designing, simulating, and building ***Thin-Film transistors*** and analysis the effects of light exposure on the device, for development of the important parameters for communication and digital applications.

PROJECTS, AND PREVIOUS RESEARCHES

- Simulation of Hermite-Gaussian and Laguerre-Gaussian laser beam profile with MATLAB using Nvidia CUDA
- Optical Transmission of 4x4 keypad to LCD with laser using two ATMega32 microcontrollers at transmitter and receiver side without involving TX and RX ports.
- Programming UDP Server and Client, using Rust language.
- Programming Multi-Threaded TCP Server and Client, using Rust language.
- A review paper on application of single photon avalanche diodes in quantum communications.
- Simulation of IEEE 802.11 physical layer for different types of channel, with interactive user interface, using MATLAB.
- Programming different schemes of quantum computing systems, using Qiskit module in Python programming language.
- Design, simulation and building RF receiver for aviation band (This project was preparation for new major of study in University of Isfahan, Avionics).
- Design, simulation and building optical circuit for tunable rare-earth doped fiber laser with new tuning technique.
- Simulation of transmission and receiving of audio signals using OFDM with MATLAB.

LANGUAGES



HOBBIES

I like learning new things, specially in my favorite field of study (Information Theory). Also, read books on philosophy of science, modern logics, and psychological novels. Plus all previous, I'm currently translating ***"Information Theory, Inference and Learning Algorithms"*** by ***"David J. C. MacKay"*** and I wish soon this translation be available, because everyone must see the beautiful teaching method of this book.

CERTIFIED BADGES